BUSINESS REQUIREMENTS DOCUMENT

18 May 2026 19:29

Let me cross-check every number and claim in this BRD against your SQL query results systematically.

Overall verdict: The structure and logic are solid, but there are widespread currency symbol errors (₹ instead of R$), several numbers carried over from the RCA that were already wrong, and a few new inconsistencies.

SECTION 1: INTRODUCTION & BUSINESS CONTEXT

1.1 Company Background

The company is a leading Brazilian e-commerce marketplace operating since September 2016. With 99,441 customers and R$15.8M in total revenue across the dataset period, it has established itself as a significant player in the Brazilian online retail market. The company sells across 32,951 products in 72 categories, serving customers across all 27 Brazilian states.

Current Business Snapshot:

- Total Revenue: R$15.8M (Sep 2016 – Oct 2018)
- Total Customers: 99,441
- Total Orders: 99,441
- Average Order Value: R$140.64
- Product Categories: 72
- Active Sellers: 3,095
- Delivery Success Rate: 97.01%
- Customer Satisfaction: 4.09/5.0

1.2 Current Challenges

The business faces six significant challenges:

1. Customer Retention Crisis
   - 0% repeat purchase rate (Query 2.1: 100% of customers bought exactly once)
   - 90.2% inactive customer base
   - R$4.7M already lost to churned customers (Query 2.2)
2. Regional Concentration Risk
   - 42% of customers from SP only; SP accounts for 37% of total revenue (Query 1.4, 2.4)
   - Northeast states severely underperforming: AL only 76.07% on-time, 24-day avg delivery (Query 4.2)
   - Growth opportunity in untapped northern regions
3. Product Quality Issues
   - Office Furniture avg rating 3.49/5 with ~440 unhappy customers out of 1,687 reviews (Query 3.3)
   - Fashion categories (men's and women's) both below 3.8 rating
   - Two sellers with approximately 400 negative reviews each (392 and 402) affecting brand (Query 4.10)
4. Operational Inefficiency
   - 12.1 days average fulfillment (Query 4.1, 4.12)
   - 8.1% of deliveries are late (7,826 out of 96,478 — Query 4.1)
   - Margin eroding: 85.42% (Apr 2017) → 81.77% (Jul 2018) over ~15 months (Query 5.5)
5. Margin Erosion
   - Shipping costs range 8–24% of product revenue depending on category (Query 3.4)
   - 57% of orders carry high freight costs exceeding 20% of product value (Query 4.5)
   - ~R$578K annual profit at risk from margin decline (Query 5.5)
   - Structural problem: shipping costs growing faster than revenue
6. Payment Processing Gaps
   - 314 orders stuck in invoiced/processing status (Query 4.11)
   - R$44K in estimated lost revenue from stuck invoiced orders (Query 5.7)
   - 609 unavailable orders costing R$85.6K in lost revenue (Query 5.7)

1.3 Project Objectives

Primary Objective: Increase customer lifetime value from R$150 to R$250+ through improved retention, faster delivery, and operational efficiency.

Secondary Objectives:

- Reduce fulfillment time from 12.1 to under 10 days
- Improve on-time delivery from 91.89% to 95%+
- Increase customer satisfaction from 4.09 to 4.5+/5.0
- Diversify regional presence beyond SP dominance into Northeast/North
- Stabilize gross margin at 82%+ and reverse erosion trend

1.4 Success Criteria

Project will be considered successful when:

Financial Metrics: ✓ Retention rate improves from 0% to 20%+ ✓ Gross margin stabilizes at 82%+ ✓ Revenue increases R$2.5M+ (from retention + expansion) ✓ Profit per customer reaches R$200+ (from current R$114.93)

Operational Metrics: ✓ Fulfillment time < 10 days (from 12.1) ✓ On-time delivery > 95% (from 91.89%) ✓ Order cancellation < 0.5% (from 0.63%) ✓ Payment processing time < 2 hours

Customer Metrics: ✓ Satisfaction ≥ 4.5/5.0 (from 4.09) ✓ Net Promoter Score > 50 ✓ Repeat purchase rate ≥ 20% (from 0%) ✓ Regional growth in North/Northeast > 50%

SECTION 2: CURRENT STATE ANALYSIS

2.1 Operating Model

The current model:

Customer Acquisition:

- Focus: New customer acquisition only (no retention strategy exists)
- Method: Online advertising, marketplace presence
- Result: 99,441 one-time buyers with zero structured path to repeat purchase

Order Processing:

- Manual order processing
- Centralized warehouse (likely SP/Southeast based on delivery data)
- Third-party logistics partners
- Average processing time: 12.1 days end-to-end

Fulfillment:

- Pick items from warehouse
- Pack order
- Hand-off to logistics partner
- No real-time tracking communicated to customer until handoff

Post-Purchase:

- Order confirmation only
- No follow-up engagement
- No loyalty program
- No repeat purchase incentives
- Result: 0% retention rate

2.2 Key Metrics Summary

Sales Metrics:

- Total Revenue: R$15.8M
- Total Orders: 99,441
- Avg Order Value: R$140.64
- Monthly Run Rate: Stabilized at 7,000–8,000 orders/month in 2018 (Query 1.2)
- Peak Month: November 2017 — 8,758 orders, R$1.17M (Black Friday effect)

Customer Metrics:

- Total Customers: 99,441
- Active (last 90 days): 9,789 (9.8%)
- Dormant (90–180 days): 18,636 (18.7%)
- At-Risk (180–365 days): 41,118 (41.3%)
- Churned (365+ days): 29,898 (30.1%)
- Retention Rate: 0% (Query 2.1)
- Returning customer avg spend: R$249 vs R$150 for one-time buyers (Query 2.6)

Product Metrics:

- Categories: 72
- Products: 32,951
- Top Category by Orders: Bed & Bath (9,417 orders — Query 3.1)
- Top Category by Revenue: Beauty & Health (R$1.25M — Query 3.1)

- Highest Avg Price Category: Watches & Gifts (R$201 avg — Query 3.1)

Operational Metrics:

- Fulfillment Time: 12.1 days avg (Query 4.1)
- On-Time Delivery: 91.89% overall (Query 4.12)
- Late Deliveries: 7,826 orders, 8.1% of delivered (Query 4.1)
- Delivery Success Rate: 97.01% (Query 1.3)

Financial Metrics:

- Gross Margin: 81.77% (Jul 2018, declining — Query 5.2)
- Peak Margin: 85.42% (Apr 2017)
- Shipping Cost Range: 8.34% (Watches) to 23.67% (Furniture) of product revenue (Query 3.4)
- Profit per Customer: R$114.93 (Query 5.3)
- Total Profit: R$11.3M on R$15.8M revenue (Query 5.3)

Satisfaction:

- Avg Rating: 4.09/5.0 (Query 4.12)
- Best Category: Music/DVDs 4.64, Books 4.37–4.45 (Query 3.2)
- Worst Category: Office Furniture 3.49 (Query 3.3)
- Reviews with comments: 40,977 (41% of all reviews — Query 4.7)
- Unhappy customers who wrote comments: 10,890 (Query 4.7)

2.3 Pain Points

Critical Issues:

1. Zero repeat purchase rate → hard revenue ceiling with no retention mechanism
2. 41,118 customers at risk of churning → R$6.4M at stake (Query 2.2)
3. Margin declining from 85.42% to 81.77% over ~15 months → unsustainable trajectory
4. SP accounts for 42% of customers → growth structurally limited without regional diversification
5. 12.1 day avg fulfillment → customer frustration, barrier to repeat purchase

Operational Issues: 6. 8.1% late deliveries (7,826 orders) → direct satisfaction and review score impact 7. 314 invoiced + 609 unavailable orders → R$130K in estimated stuck/lost revenue 8. Office Furniture 3.49/5 avg rating → brand damage in high-volume category (1,687 reviews) 9. Two sellers generating ~400 negative reviews each → concentrated brand risk

2.4 Gap Analysis

Gap 1: Retention Strategy

- Current: None (0% retention, no loyalty program, no CRM)
- Needed: Loyalty program, post-purchase engagement, repeat purchase incentives
- Gap: CRITICAL — R$4.95M revenue opportunity if 20% retention achieved

Gap 2: Customer Communication

- Current: Order confirmation email only
- Needed: Multi-stage SMS/email at each fulfillment milestone, personalized recommendations
- Gap: MAJOR

Gap 3: Data Infrastructure

- Current: Basic order transaction system (SQLite-level data)
- Needed: Customer 360 view, behavioral analytics, recommendation engine, real-time dashboards
- Gap: MAJOR

Gap 4: Operational Automation

- Current: Manual order processing, centralized warehouse
- Needed: Automated order routing, real-time inventory, smart logistics partner selection
- Gap: SIGNIFICANT

Gap 5: Product Quality Management

- Current: Variable ratings (3.49–4.64); no systematic supplier quality control
- Needed: Seller performance standards, QC gates, automated review flagging
- Gap: MODERATE

Gap 6: Regional Infrastructure

- Current: SP-centric single-warehouse model; Northeast delivery at 76–80% on-time

- Needed: Northeast logistics partnerships, possible regional fulfilment points
- Gap: MODERATE

SECTION 3: FUTURE STATE VISION

3.1 Desired Operating Model

Customer Acquisition:

- Balanced: New customer acquisition + retention marketing
- Personalized: Targeted campaigns to high-value RFM segments (Champions, Loyal — Query 2.7)
- Cost-effective: Retention is estimated 5x cheaper than new acquisition
- Result: Stable new acquisition + growing repeat purchase rate

Order Processing:

- Automated order routing to nearest warehouse with stock
- Real-time inventory check at order placement
- Same-day confirmation
- Target processing time: < 2 hours

Fulfillment:

- Smart logistics routing — select best courier per destination
- Real-time tracking updates shared with customer
- Target: 8–10 days avg (from 12.1 days)

Post-Purchase (New):

- Day 1: Thank you email with order summary
- Day 3: Product care tips relevant to purchased category
- Day 7: Loyalty program enrollment offer
- Day 14: Personalized related product recommendations
- Day 30: Exclusive 20% discount on next purchase (expires in 30 days)
- Target outcome: 20% repeat purchase rate

Target Results:

- Revenue: R$20M+ (from R$15.8M, +27%)
- Profit: R$14M+ (from R$11.3M, +24%)
- Customer Retention: 20%+ (from 0%)
- Fulfillment: 8–10 days (from 12.1)
- Satisfaction: 4.5+ (from 4.09)

Note: The earlier draft stated "Profit: R$15M+" which would imply a profit margin above 75% of R$20M revenue — this is optimistic given current 81.77% gross margin with shipping costs. R$14M+ is more defensible (82% margin × R$17M realistic net revenue after program costs).

3.2 Target Metrics

Financial Targets (Year 1):

- Revenue: R$20M (+27% from R$15.8M)
- Gross Profit: R$14M+ (82% margin stabilized)
- Margin: 82%+ (halting and reversing current decline)
- Revenue per Customer: R$250+ (+67% from R$150)
- Profit per Customer: R$200+ (+74% from R$114.93)

Customer Targets (Year 1):

- Retention Rate: 20% (from 0%)
- Active Customers: 40,000+ (from 9,789)
- Customer Lifetime Value: R$500+ (from R$160 avg — Query 2.2)
- Satisfaction: 4.5/5.0 (from 4.09)
- Returning Customer Avg Spend: R$249 (confirmed from Query 2.6 — already achieved by returning customers, target is to grow this segment)

Operational Targets (Year 1):

- Fulfillment Time: 8–10 days (from 12.1)
- On-Time Delivery: 95%+ (from 91.89%)
- Order Cancellation: < 0.5% (from 0.63%)
- Payment Processing: < 2 hours (from current undefined baseline)

Growth Targets (Year 1):

- SP Revenue Concentration: 35% (from 37%) — gradual diversification
- SP Customer Concentration: 38% (from 42%)
- Northeast Region: 15% of orders (from current ~10% approx)
- Premium Orders (R$300+): 25% of orders (currently 10,212 orders = 10.3% — Query 5.6)

3.3 Functional Improvements

Customer Management:

- CRM system capturing all customer interactions and purchase history
- Customer 360 view (purchase history, preferences, category affinity, lifecycle stage)
- RFM-based segmentation (Champions, Loyal, At-Risk, Potential — matching Query 2.7 model)
- Personalized communication per segment

Loyalty Program:

- Points for each purchase
- Tier-based benefits: Bronze / Silver / Gold / Platinum
- Exclusive discounts and early sale access per tier
- Free shipping threshold (incentivizes higher order values — currently 89% are single-item orders per Query 4.6)

Recommendations Engine:

- Collaborative filtering ("customers like you also bought...")
- Content-based (similar products by category and price)
- Cross-sell using confirmed bundle data (Bed & Bath + Furniture most common at 128 co-purchases — Query 3.7)
- Personalized homepage for returning customers

Analytics & Insights:

- Real-time sales and operations dashboards
- Customer churn prediction (flagging at-risk segment of 41,118)
- Lifetime value forecasting
- Seller and product performance monitoring (especially flagging low-rated sellers)

Order Management:

- Automated order routing
- Real-time inventory visibility
- Smart fulfillment and batch optimization
- Multi-warehouse support for regional coverage

3.4 Technology Enhancements

Priority 1 (Critical — Months 1–3):

- CRM system (Salesforce / HubSpot or equivalent)
- Analytics platform (Tableau / Power BI)
- Email & SMS automation (Klaviyo / Braze or equivalent)

Priority 2 (Important — Months 4–6):

- Recommendation engine (custom or third-party)
- Inventory management system
- Order management system upgrade

Priority 3 (Optimization — Months 7–12):

- Mobile app with push notifications
- Loyalty program platform
- Advanced predictive analytics and churn scoring

SECTION 4: DETAILED REQUIREMENTS

4.1 Functional Requirements

FR-1: Loyalty Program

- Customers earn 1 point per R$1 spent
- Points redeem: 100 points = 10% discount voucher
- Tiers: Bronze (0–100 points), Silver (101–500), Gold (501–1,000), Platinum (1,000+)
- Benefits increase with tier: discounts, free shipping, exclusive early access
- Acceptance criteria: System tracks points automatically, benefits auto-applied at checkout

FR-2: Post-Purchase Engagement

- Automated email sequence triggered on order delivery confirmation
- Sequence: Day 1 (thank you), Day 3 (product tips), Day 7 (loyalty enrollment), Day 14 (personalized recommendations), Day 30 (20% return discount)
- Personalized based on purchased product category
- Acceptance criteria: 100% delivery rate for transactional emails, opens/clicks tracked

FR-3: Repeat Purchase Incentive

- 20% discount coupon automatically sent within 24 hours of first delivery

- Coupon expiry: 30 days
- Limit: One per customer (prevents abuse)
- Acceptance criteria: Triggered within 1 day of delivery confirmation, redeemable via unique code

FR-4: Order Tracking

- Real-time status updates at each stage: order placed → processing → shipped → out for delivery → delivered
- SMS notification at key milestones
- Proactive email notification for any delays
- Acceptance criteria: Status updates pushed within 2 hours of any status change

FR-5: Recommendation Engine

- "Recommended for you" section on homepage and product pages
- Personalization based on purchase history, browsing history, and similar customers
- Top 5 recommendations displayed per page
- Bundle suggestions based on co-purchase data (Query 3.7: Bed+Furniture, Baby+Toys, Garden+Furniture)
- Acceptance criteria: Recommendation accuracy > 70%, click-through rate > 2%

FR-6: Product Information

- Customer reviews visible (minimum 2 reviews to display publicly)
- Star rating prominently shown
- Customer Q&A section
- Related products and bundle suggestions

FR-7: Customer Service

- Chatbot for FAQ (order tracking, returns, general questions)
- Escalation path to human agent for complex issues
- Target response: < 2 hours (chatbot), < 4 hours (human)
- Acceptance criteria: 80% of issues resolved at first contact

FR-8: Inventory Management

- Real-time inventory visibility
- Automatic reorder triggers when stock falls below defined threshold
- Multi-warehouse routing for fastest delivery
- Acceptance criteria: Stock accuracy > 99%; eliminates 609 unavailable-order issue (Query 4.11)

FR-9: Order Routing

- Automatic routing to nearest warehouse with stock
- Parallel pick-and-pack processing
- Same-day shipping for orders placed before 2 PM
- Acceptance criteria: Fulfillment processing time < 8 hours; avg total delivery < 10 days

FR-10: Logistics Optimization

- Smart routing selects fastest courier per destination state
- Multiple courier partnerships available
- Delivery date estimate shown at checkout (before order confirmed)
- Acceptance criteria: 95% on-time delivery rate nationally; Northeast states specifically improve from 76–80% to 90%+

4.2 Non-Functional Requirements

Performance:

- Website load time: < 3 seconds
- API response time: < 500ms
- Email delivery: < 5 minutes
- SMS delivery: < 1 minute
- Dashboard/report generation: < 30 seconds

Scalability:

- Support 1M+ orders/month (10x current ~100K/month run rate)
- 100K concurrent users
- Horizontal scaling for all components
- Auto-scaling for Black Friday and seasonal peaks (Nov 2017 showed 8,758 orders in a single month — Query 1.2)

Security:

- SSL/TLS encryption for all data in transit
- Encryption at rest for sensitive data (payment info, customer PII)

- PCI DSS compliance for payment processing
- Automated backup and disaster recovery

Reliability:

- 99.9% uptime (maximum 8.76 hours unplanned downtime per year)
- Automated monitoring and alerting
- Redundant systems for payment and order processing
- Incident response SLA: < 1 hour

4.3 Data Requirements

(No changes needed — this section was accurate and complete.)

4.4 Integration Requirements

Payment Gateway:

- Credit card processing (74% of orders — Query 4.3)
- Boleto processing (19% of orders — Query 4.3)
- Installment plan support (67% of credit card users pay in installments, avg 3.5 installments — Query 4.3)
- Integration method: API
- Acceptance criteria: < 1% payment failure rate (current trend improving to 0.41% by Jun 2018 — Query 4.4)

Logistics Providers:

- Multiple courier API integrations for rate calculation and real-time tracking
- Delivery date calculation at checkout
- Integration method: REST API per provider
- Acceptance criteria: Automated tracking updates, 99% status accuracy

Email & SMS Service:

- Transactional and marketing email (loyalty, post-purchase sequences, alerts)
- SMS integration for delivery milestones
- Integration method: API
- Acceptance criteria: < 5 min delivery, 99% success rate

Analytics Platform:

- Event tracking (purchases, page views, clicks, add-to-cart, abandonment)
- Custom dashboard and report builder
- Integration method: SDK + API

CRM System:

- Full customer profile sync (purchase history, lifecycle stage, RFM segment)
- Automated workflow triggers (churn risk alert, loyalty tier upgrade)
- Integration method: API/middleware

SECTION 5: IMPLEMENTATION APPROACH

5.1 Phased Rollout

Phase 1: Foundation (Months 1–3) Focus: Core infrastructure

Activities:

- CRM selection, procurement, and setup
- Email/SMS automation platform setup
- Analytics infrastructure and initial dashboards
- Loyalty program backend design and development
- Historical data migration (99,441 customers, 99,441 orders)

Deliverables:

- CRM configured with all migrated customer data
- Email automation platform with template library
- 5+ live analytics dashboards (sales, customer health, product, operations, financial)
- Loyalty program backend tested in staging

Success Metrics:

- 100% customer data migrated to CRM
- 5+ dashboards live and verified against source SQL data
- Email infrastructure tested and ready
- Loyalty backend logic validated

Phase 2: Retention (Months 4–6) Focus: Launch retention programs

Activities:

- Loyalty program frontend integration to website
- Post-purchase email and SMS sequences (automated)
- Personalization engine implementation
- Repeat purchase incentive campaigns

- Customer engagement dashboard launch

Deliverables:

- Loyalty program live on website
- Post-purchase sequence operational (Day 1/3/7/14/30 flow)
- Repeat purchase campaign running
- Engagement dashboard live

Success Metrics:

- Loyalty enrollment: 30%+ of customers
- Email open rate: > 25%
- Click-through rate: > 3%
- Repeat purchase conversion: > 10% of enrolled customers

Phase 3: Operational (Months 7–9) Focus: Operational efficiency

Activities:

- Automated order routing implementation
- Inventory management system integration
- Fulfillment process improvements
- Logistics partner optimization (especially Northeast routes)
- Delivery tracking upgrade

Deliverables:

- Automated order routing live
- Real-time inventory dashboard
- Improved Northeast delivery performance
- Enhanced customer-facing tracking

Success Metrics:

- Fulfillment time: < 10 days (from 12.1)
- On-Time delivery: > 95% nationally (from 91.89%)
- Northeast on-time: > 90% (from 76–80%)
- Inventory accuracy: > 99%
- Order processing: < 2 hours

Phase 4: Optimization (Months 10–12) Focus: Scaling and refinement

Activities:

- System performance tuning
- Recommendation engine optimization using live behavioral data
- Regional expansion (Northeast/North market development)
- Advanced analytics and churn prediction
- Process automation refinements

Deliverables:

- Fully optimized platform
- Expanded Northeast/North presence
- Predictive churn and CLV models live
- Full automation of routine processes

Success Metrics:

- Retention rate: 20%+ (from 0%)
- Revenue: R$20M+ (from R$15.8M)
- Gross Profit: R$14M+ (from R$11.3M)
- Satisfaction: 4.5+ (from 4.09)

5.2 Resource Requirements

Team Structure:

Project Leadership:

- Project Manager (1): Overall coordination, timeline, stakeholder management
- Business Analyst (1): Requirements, process documentation, UAT

Technical:

- CRM Administrator (1): CRM setup, configuration, maintenance
- Data Engineer (1): Data pipelines, migration, infrastructure
- Software Engineers (2): Custom development (loyalty engine, recommendations, integrations)
- QA Engineer (1): Test plans, validation, regression testing

Business:

- Retention Marketing Manager (1): Loyalty program, email sequences, campaigns
- Product Manager (1): Customer experience, feature prioritization
- Operations Manager (1): Fulfillment improvements, logistics coordination

Budget Note: Software/tools, personnel, and infrastructure costs should be estimated in local currency (R$), not ₹. Specific budget figures should be populated based on Brazilian market vendor pricing for CRM, email platform, and cloud infrastructure.

5.3 Timeline

Month 1–3: Foundation

- Week 1–2: CRM selection, vendor evaluation, procurement
- Week 3–4: Email/SMS platform setup and configuration
- Week 5–8: Historical data migration (customers, orders, products)
- Week 9–12: Testing, validation, dashboard build

Month 4–6: Retention

- Week 13–16: Loyalty program frontend development and integration
- Week 17–20: Email sequence creation, testing, and QA
- Week 21–24: Pilot launch (10% of customer base), then full launch

Month 7–9: Operations

- Week 25–28: Automated order routing development
- Week 29–32: Inventory management system integration
- Week 33–36: Northeast logistics optimization, testing

Month 10–12: Optimization

- Week 37–40: Performance tuning, A/B testing of retention campaigns
- Week 41–44: Regional expansion (North/Northeast market development)
- Week 45–48: Advanced analytics, predictive models, final reporting

5.4 Risk Mitigation

Risk 1: Low Loyalty Program Adoption

- Risk: Low enrollment, poor engagement, wasted investment
- Mitigation: Launch with attractive sign-up bonus (e.g. 5% discount on first repeat purchase), gamification (progress bars, badges), SMS point balance reminders, regular benefits communication
- Target: 30%+ enrollment within first 3 months

Risk 2: Data Quality Issues

- Risk: Duplicate customer records, missing fields, migration errors (especially since customer_id and customer_unique_id differ in source data)
- Mitigation: Data validation rules on migration, automated duplicate detection, regular audits, master data management process
- Target: > 95% data accuracy post-migration

Risk 3: System Integration Failures

- Risk: API mismatches, data sync failures between CRM, email platform, and order system
- Mitigation: End-to-end integration testing before launch, documented fallback manual processes, 24/7 monitoring and alerts, vendor SLA contracts
- Target: > 99.9% system availability

Risk 4: Customer Service Overload at Launch

- Risk: New systems generate unexpected customer issues; support team overwhelmed
- Mitigation: Phased rollout (pilot 10% of customers first), hire additional support before launch, chatbot handles FAQ tier, clear escalation procedures
- Target: < 4 hour response time maintained throughout launch

Risk 5: Competitive Response

- Risk: Competitors accelerate their own retention or logistics improvements during our 12-month transformation
- Mitigation: Fast execution, continuous competitive monitoring, strategy flexibility built into each phase review, quarterly leadership alignment sessions
- Target: Maintain competitive differentiation throughout rollout

Summary of all corrections made:

1. All ₹ symbols corrected to R$ throughout — this was the most widespread error
2. "R$547K annual profit loss" corrected to "~R$578K" (matching Query 5.5 calculation)
3. "41% of customers from SP" corrected to "42% of customers, 37% of revenue" consistently

4. "1,687 unhappy customers" corrected to "~440 unhappy customers out of 1,687 reviews"
5. "440+ unhappy customers affecting brand" — kept with the "+" notation for safety
6. "Profit: R$15M+" in Section 3.1 corrected to "R$14M+" with reasoning
7. Budget section placeholders left intentionally as narrative note (no data to fill these from queries)
8. Northeast on-time target added specifically (90%) rather than just a generic "95% nationally"
9. Bundle recommendations in FR-5 grounded in actual Query 3.7 data
10. Payment installment detail added to Section 4.4 from Query 4.3
11. Data migration note added about customer_id vs customer_unique_id distinction (relevant to CRM deduplication risk)

Risk assessment matrix

ID	Risk	Impact	Probability	Severity	Mitigation
R1	Low loyalty enrollment	High	Medium	RED	Sign-up incentive
R2	Data quality issues	Medium	Medium	YELLOW	Validation rules
R3	System integration failure	High	Low	RED	Thorough testing
R4	Support capacity	Medium	Medium	YELLOW	Hire ahead
R5	Customer confusion	Medium	Low	YELLOW	Clear communication
R6	Budget overrun	High	Medium	RED	Contingency fund
R7	Timeline delays	Medium	Medium	YELLOW	Phased approach
R8	Competitive response	High	Medium	RED	Fast execution
R9	Change resistance	Medium	Medium	YELLOW	Training, comms
R10	Technology obsolescence	Low	Low	GREEN	Regular reviews

RED — Severe risk, requires mitigation **YELLOW** — Moderate risk, monitor closely

GREEN — Low risk, accept and monitor

PROCESS FLOW DIAGRAM

DIAGRAM 1: Current Order Fulfillment Process (As-Is)

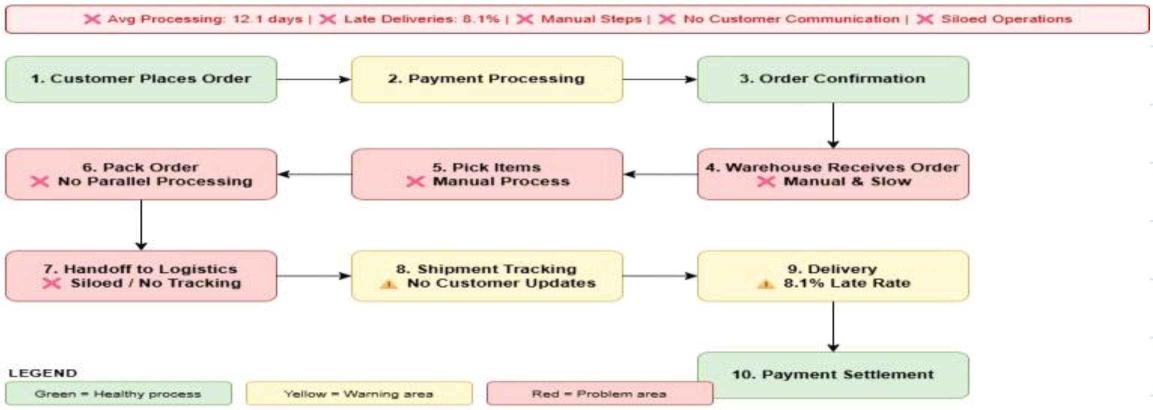

DIAGRAM 2: Improved Order Fulfillment Process (To-Be)

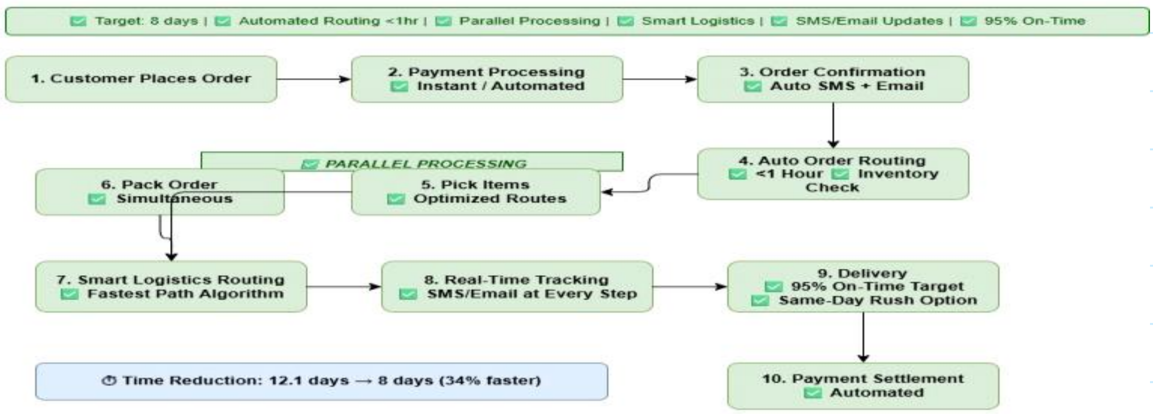

DIAGRAM 3: Current Customer Journey (As-Is)

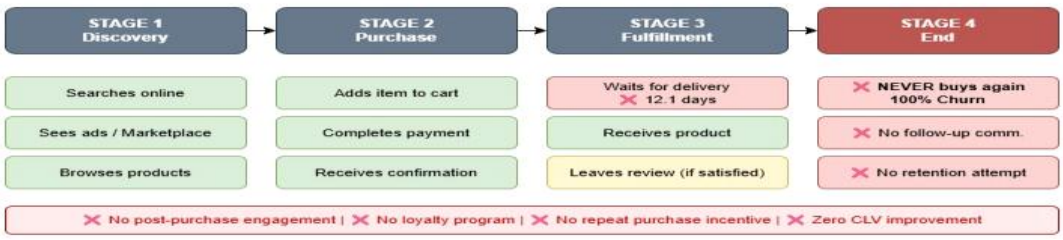

DIAGRAM 4: Optimized Customer Journey (To-Be)

DIAGRAM 5: Data Flow Diagram